

Chapter 1



Data Analysis

Section 1.2

Displaying Quantitative Data with Graphs

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Displaying Quantitative Data with Graphs

LEARNING TARGETS

By the end of this section, you should be able to:

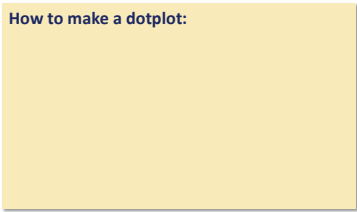
- ✓MAKE and INTERPRET dotplots, stemplots, and histograms of quantitative data.
- ✓IDENTIFY the shape of a distribution from a graph.
- ✓DESCRIBE the overall pattern (shape, center, and variability) of a distribution and IDENTIFY any major departures from the pattern (outliers).
- ✓COMPARE distributions of quantitative data using dotplots, stemplots, and histograms.

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Dotplots

A dotplot shows each data value as a dot above its location on a number line.

How to make a dotplot:



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Dotplots

A dotplot shows each data value as a dot above its location on a number line.

How to make a dotplot:

- 1) Draw a horizontal axis (a number line) and label it with the quantitative variable.



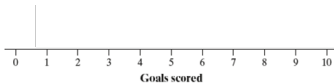
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Dotplots

A dotplot shows each data value as a dot above its location on a number line.

How to make a dotplot:

- 1) Draw a horizontal axis (a number line) and label it with the quantitative variable.
- 2) Scale the axis from the minimum to the maximum value.



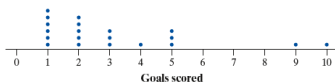
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Dotplots

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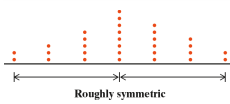
How to make a dotplot:

- 1) Draw a horizontal axis (a number line) and label it with the quantitative variable.
- 2) Scale the axis from the minimum to the maximum value.
- 3) Mark a dot above the location on the horizontal axis corresponding to each data value.



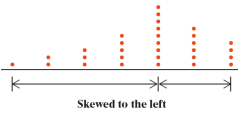
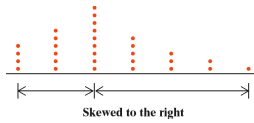
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Describing Shape



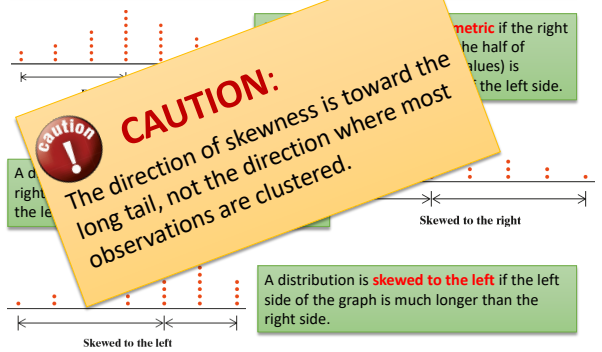
A distribution is roughly **symmetric** if the right side of the graph (containing the half of observations with the largest values) is approximately a mirror image of the left side.

A distribution is **skewed to the right** if the right side of the graph is much longer than the left side.

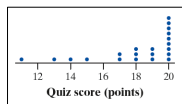


A distribution is **skewed to the left** if the left side of the graph is much longer than the right side.

Describing Shape

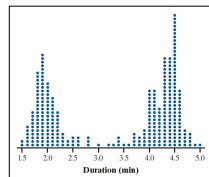
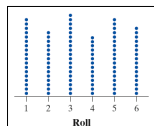


Describing Shape



The distribution of a quantitative variable is **unimodal** if it has a single peak.

The distribution of a quantitative variable is **bimodal** if it has two distinct clusters and peaks.



The distribution of a quantitative variable is **approximately symmetric** if the frequencies are about the same for all values.

Describing Distributions

HOW TO DESCRIBE THE DISTRIBUTION OF A QUANTITATIVE VARIABLE

In any graph, look for the *overall pattern* and for clear *departures* from that pattern.

- You can describe the overall pattern of a distribution by its shape, center, and variability.
- An important kind of departure is an outlier, an observation that falls outside the overall pattern.

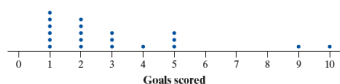
AP® Exam Tip

Always be sure to include context when you are asked to describe a distribution. This means using the variable name, not just the units the variable is measured in.

Describing Distributions

Describe the distribution of goals scored in 20 games played by the 2016 U.S. women's soccer team.

Did we include context?



Shape: The distribution of goals scored is skewed to the right, with a single peak at 1 goal. There is a gap between 5 and 9 goals.

Outliers: The games when the team scored 9 and 10 goals appear to be outliers.

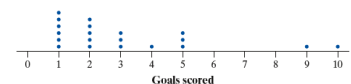
Center: The median is 2 goals scored.

Variability: The number of goals varies from 1 to 10 goals scored.

Describing Distributions

Describe the distribution of goals scored in 20 games played by the 2016 U.S. women's soccer team.

Did we include context? **YES!**



Shape: The distribution of **goals scored** is skewed to the right, with a single peak at 1 **goal**. There is a gap between 5 and 9 **goals**.

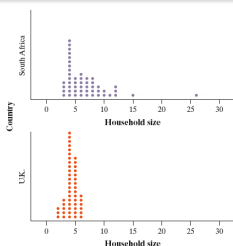
Outliers: The **games** when the **team** scored 9 and 10 **goals** appear to be outliers.

Center: The median is 2 **goals scored**.

Variability: The **number of goals** varies from 1 to 10 **goals scored**.

Comparing Distributions

We used Census At School's "Random Data Selector" to choose 50 students from each country. Here are dotplots of the household sizes reported by the survey respondents. Compare the distributions of household size for these two countries.

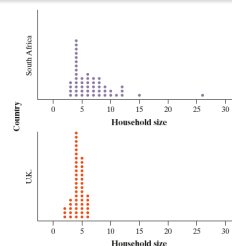


AP® Exam Tip

When comparing distributions of quantitative data, it's not enough just to list values for the center and variability of each distribution. You must explicitly compare these values, using words like "greater than," "less than," or "about the same as."

Comparing Distributions

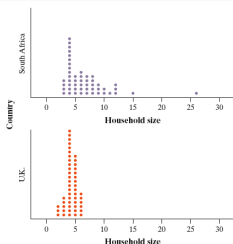
Shape: The distribution of household size for the U.K. sample is roughly symmetric, with a single peak at 4 people. The distribution of household size for the South Africa sample is skewed to the right, with a single peak at 4 people and a clear gap between 15 and 26.



Comparing Distributions

Shape: The distribution of household size for the U.K. sample is roughly symmetric, with a single peak at 4 people. The distribution of household size for the South Africa sample is skewed to the right, with a single peak at 4 people and a clear gap between 15 and 26.

Outliers: There don't appear to be any outliers in the U.K. distribution. The South African distribution seems to have two outliers: the households with 15 and 26 people.

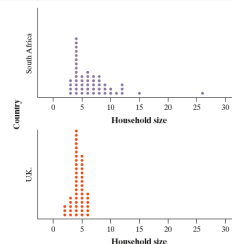


Comparing Distributions

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Outliers: There don't appear to be any outliers in the U.K. distribution. The South African distribution seems to have two outliers: the households with 15 and 26 people.

Center: Household sizes for the South African students tend to be larger (median 6 people) than for the U.K. students (median 4 people).



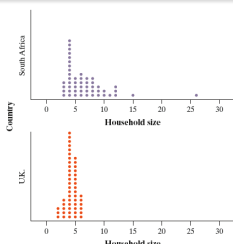
Comparing Distributions

Shape: The distribution of household size for the U.K. sample is roughly symmetric, with a single peak at 4 people. The distribution of household size for the South Africa sample is skewed to the right, with a single peak at 4 people and a clear gap between 15 and 26.

Outliers: There don't appear to be any outliers in the U.K. distribution. The South African distribution seems to have two outliers: the households with 15 and 26 people.

Center: Household sizes for the South African students tend to be larger (median 6 people) than for the U.K. students (median 4 people).

Variability: The household sizes for the South African students vary more (from 3 to 26 people) than for the U.K. students (from 2 to 6 people).



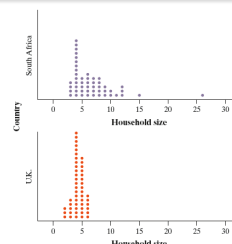
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Variability: The household sizes for the South African students vary more (from 3 to 26 people) than for the U.K. students (from 2 to 6 people).



✓ Context
✓ Comparative language

Stemplots

A **stemplot** shows each data value separated into two parts: a stem, which consists of all but the final digit, and a leaf, the final digit. The stems are ordered from lowest to highest and arranged in a vertical column. The leaves are arranged in increasing order out from the appropriate stems.

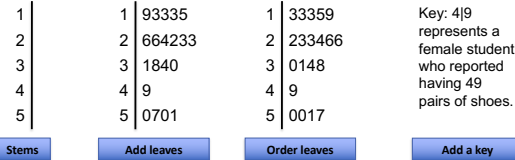
How to make a stemplot:

- 1) Separate each observation into a stem, consisting of all but the final digit, and a leaf, the final digit. Write the stems in a vertical column with the smallest at the top. Draw a vertical line at the right of this column.
- 2) Write each leaf in the row to the right of its stem.
- 3) Arrange the leaves in increasing order out from the stem.
- 4) Provide a key that identifies the variable and explains what the stems and leaves represent.

Stemplots

These data represent the responses of 20 female AP Statistics students to the question, "How many pairs of shoes do you have?" Construct a stemplot.

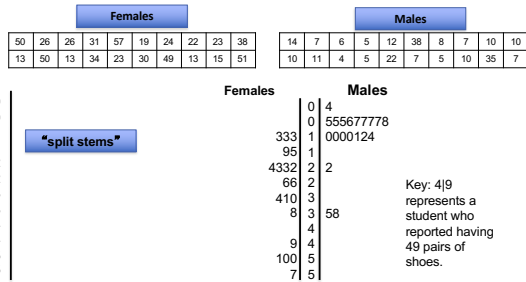
50	26	26	31	57	19	24	22	23	38
13	50	13	34	23	30	49	13	15	51



Stemplots

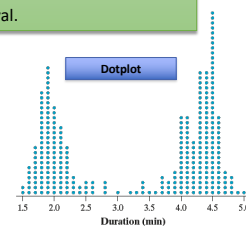
When data values are "bunched up", we can get a better picture of the distribution by **splitting stems**.

Two distributions of the same quantitative variable can be compared using a **back-to-back stemplot** with common stems.



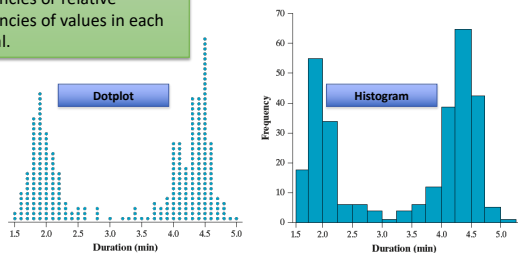
Histograms

A **histogram** shows each interval of values as a bar. The heights of the bars show the frequencies or relative frequencies of values in each interval.



Histograms

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Histograms

How to make a histogram:

State	Percent	State	Percent	State	Percent
Alabama	2.8	Louisiana	2.9	Ohio	3.6
Alaska	7.0	Maine	3.2	Oklahoma	4.9
Arizona	15.1	Maryland	12.2	Oregon	9.7
Arkansas	3.8	Massachusetts	14.1	Pennsylvania	5.1
California	27.2	Michigan	5.9	Rhode Island	12.9
Colorado	10.3	Minnesota	6.6	South Dakota	4.1
Connecticut	12.9	Mississippi	1.9	Tennessee	3.9
Delaware	5.1	Missouri	3.3	Texas	15.9
Florida	15.9	Montana	1.9	Utah	8.3
Georgia	9.2	Nevada	5.6	Vermont	3.9
Hawaii	16.3	New Hampshire	5.4	Virginia	10.1
Idaho	5.6	New Jersey	20.1	Washington	12.4
Illinois	13.8	New Mexico	10.1	West Virginia	1.2
Indiana	4.2	New York	21.6	Wisconsin	4.4
Iowa	3.8	North Carolina	6.9	Wyoming	2.7
Kansas	6.3	North Dakota	2.1		
Kentucky	2.7				

Histograms

How to make a histogram:

- 1) Choose equal-width intervals that span the data.

Frequency Table

Class	
0 to <5	
5 to <10	
10 to <15	
15 to <20	
20 to <25	
25 to <30	
Total	

State	Percent	State	Percent	State	Percent
Alabama	2.8	Louisiana	2.9	Ohio	3.6
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Colorado	10.3	Minnesota	6.6	South Carolina	4.1
Connecticut	12.9	Mississippi	1.9	South Dakota	2.2
Delaware	6.1	Missouri	3.3	Tennessee	3.9
Florida	18.9	Montana	1.9	Texas	15.9
Georgia	9.2	Nebraska	5.6	Utah	8.3
Hawaii	16.3	Nevada	19.1	Vermont	3.9
Idaho	5.6	New Hampshire	5.4	Virginia	10.1
Illinois	13.8	New Jersey	20.1	Washington	12.4
Indiana	4.2	New Mexico	10.1	West Virginia	1.2
Iowa	3.8	New York	21.6	Wisconsin	4.4
Kansas	6.3	North Carolina	6.9	Wyoming	2.7
Kentucky	2.7	North Dakota	2.1		

Histograms

How to make a histogram:

- 1) Choose equal-width intervals that span the data.
- 2) Make a table that shows the frequency or relative frequency of individuals in each interval.

Frequency Table

Class	Count
0 to <5	20
5 to <10	13
10 to <15	9
15 to <20	5
20 to <25	2
25 to <30	1
Total	50

State	Percent	State	Percent	State	Percent
Alabama	2.8	Louisiana	2.9	Ohio	3.6
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Arizona	15.1	Maryland	12.2	Oregon	9.7
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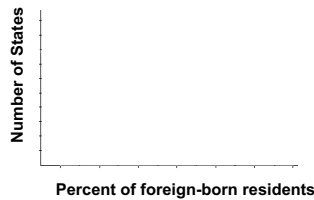
Histograms

How to make a histogram:

- 1) Choose equal-width intervals that span the data.
- 2) Make a table that shows the frequency or relative frequency of individuals in each interval.
- 3) Draw horizontal and vertical axes. Label the axes.

Frequency Table

Class	Count
0 to <5	20
5 to <10	13
10 to <15	9
15 to <20	5
20 to <25	2
25 to <30	1
Total	50



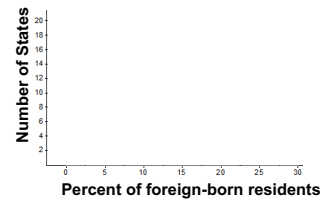
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How to make a histogram:

- 1) Choose equal-width intervals that span the data.
- 2) Make a table that shows the frequency or relative frequency of individuals in each interval.
- 3) Draw horizontal and vertical axes. Label the axes.
- 4) Scale the axes.

Frequency Table

Class	Count
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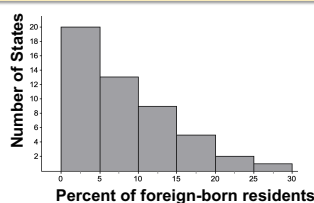
Histograms

How to make a histogram:

- 1) Choose equal-width intervals that span the data.
- 2) Make a table that shows the frequency or relative frequency of individuals in each interval.
- 3) Draw horizontal and vertical axes. Label the axes.
- 4) Scale the axes.
- 5) Draw bars above the intervals. The bar heights correspond to the frequency or relative frequency of individuals in that interval.

Frequency Table

Class	Count
0 to <5	20
5 to <10	13
10 to <15	9
15 to <20	5
20 to <25	2
25 to <30	1
Total	50



Histograms

CAUTION:

- 1) Don't confuse histograms and bar graphs.
- 2) Use percents or proportions instead of counts on the vertical axis when comparing distributions with different numbers of observations.
- 3) Just because a graph looks nice doesn't make it a meaningful display of data.

Section Summary

LEARNING TARGETS

After this section, you should be able to:

- ✓ **MAKE** and **INTERPRET** dotplots, stemplots, and histograms of quantitative data.
- ✓ **IDENTIFY** the shape of a distribution from a graph.
- ✓ **DESCRIBE** the overall pattern (shape, center, and variability) of a distribution and **IDENTIFY** any major departures from the pattern (outliers).
- ✓ **COMPARE** distributions of quantitative data using dotplots, stemplots, and histograms.